

$(\sigma E[x]) \longrightarrow [\text{var}]$
 $(\sigma E[v]) \longrightarrow$ where $v = \text{lookup}[\sigma, x]$
 $(\sigma E[(\text{if } \#false\ e_1\ e_2)]) \longrightarrow [\text{if-false}]$
 $(\sigma E[e_2])$
 $(\sigma E[(\text{if } v\ e_1\ e_2)]) \longrightarrow [\text{if-true}]$
 $(\sigma E[e_1])$ where $(\text{not } (\text{equal? } \#false\ v))$
 $(\sigma_0 E[(\text{set! } x\ v)]) \longrightarrow [\text{set!}]$
 $(\sigma_1 E[(\text{void})])$ where $\sigma_1 = \text{ext1}[\sigma_0, (x\ v)]$
 $(\sigma E[(\text{fix } v_{rec})\ v_{arg}\ \dots]) \longrightarrow [\text{fix}]$
 $(\sigma E[(v_{rec}\ (\text{fix } v_{rec}))\ v_{arg}\ \dots])$ where $(\text{lambda } (x\ \dots)\ e) = v_{rec}$
 $(\sigma_0 E[(\text{lambda } (x\ \dots_1)\ e)\ v\ \dots_1]) \longrightarrow [\text{app}]$
 $(\sigma_1 E[e])$ where $\sigma_1 = \text{ext}[\sigma_0, (x\ v), \dots]$
 $(\sigma_0 E[(\text{let } ([x\ v]\ \dots)\ e_{body})]) \longrightarrow [\text{let}]$
 $(\sigma_1 E[e_{body}])$ where $\sigma_1 = \text{ext}[\sigma_0, (x\ v), \dots]$
 $(\sigma E[(\text{slot } x_{field}\ v_{struct})]) \longrightarrow [\text{slot}]$
 $(\sigma E[v])$ where $v = \text{lookup}[(x_s\ v_s)\ \dots, x_{field}]$
 $(\sigma_0 \text{ prog}) \longrightarrow [\text{box}]$
 $(\sigma_1 E[(\text{ptr } x)])$ where $\sigma_1 = \text{ext1}[\sigma_0, (x\ v)], \text{prog} = E[(\text{box } v)]$
 $(\sigma E[(\text{unbox } v)]) \longrightarrow [\text{unbox}]$
 $(\sigma E[v_{unboxed}])$
 $(\sigma_0 E[(\text{set-box! } v_{ptr}\ v_{new})]) \longrightarrow [\text{set-box!}]$
 $(\sigma_1 E[(\text{void})])$ where $\sigma_1 = \text{ext1}[\sigma_0, (x_{addr}\ v_{new})]$
 $(\sigma E[(+ \text{ number } \dots)]) \longrightarrow [\text{add}]$
 $(\sigma E[(\text{apply } +\ (\text{number } \dots))])$
 $(\sigma E[(- \text{ number } \dots)]) \longrightarrow [\text{subtract}]$
 $(\sigma E[(\text{apply } -\ (\text{number } \dots))])$
 $(\sigma E[(\text{num->string } \text{ number})]) \longrightarrow [\text{num->string}]$
 $(\sigma E[(\text{number->string } \text{ number})])$
 $(\sigma E[(= \text{ number } \dots)]) \longrightarrow [\text{num=}]$
 $(\sigma E[(\text{apply } =\ (\text{number } \dots))])$
 $(\sigma E[(< \text{ number } \dots)]) \longrightarrow [\text{num<}]$
 $(\sigma E[(\text{apply } <\ (\text{number } \dots))])$
 $(\sigma E[(> \text{ number } \dots)]) \longrightarrow [\text{num>}]$
 $(\sigma E[(\text{apply } >\ (\text{number } \dots))])$
 $(\sigma E[(\leq \text{ number } \dots)]) \longrightarrow [\text{num<=}]$
 $(\sigma E[(\text{apply } \leq\ (\text{number } \dots))])$
 $(\sigma E[(\geq \text{ number } \dots)]) \longrightarrow [\text{num>=}]$
 $(\sigma E[(\text{apply } \geq\ (\text{number } \dots))])$
 $(\sigma E[(\text{append string } \dots)]) \longrightarrow [\text{append}]$
 $(\sigma E[(\text{apply string-append } (\text{string } \dots))])$